Pro Football Sports Science Research Update

February 2024

A practical guide to improve sleep and performance in athletes. International Journal of Sports Science & Coaching, Ahead of Print: 1-12, 2023.

Ammonia Inhalants: Use, misuse, and role in sports performance. Sports Health, Ahead of Print: 1-5, 2023.

Are return-to-play times longer in lower-limb muscle injuries involving the intramuscular tendon? A systematic review. Journal of Science and Medicine in Sport. Ahead of Print: 1-11, 2023.

Associations between cognitive function and ACL injury-related biomechanics: A systematic review. Sports Health, Vol. 15(6): 855-866, 2023.

Athletes can benefit from increased intake of EPA and DHA – Evaluating the evidence. Nutrients, 15: 4925, 2023.

Blood and MRI biomarkers of mild traumatic brain injury in non-concussed collegiate football players. Scientific Reports, 14: 665, 2024.

Changes in health-related parameters associated with sports performance drugs. International Journal of Sports Medicine, 44(3): 206-214, 2023.

Decreased concussion incidence following the implementation of the Targeting Rules: An updated epidemiology of National Football League concussions from 2017 to 2022. Cureus, 15(12): e50997, 2023.

Does increased core temperature alter cognitive performance during exercise-induced heat strain? A narrative review. 135: 35-52, 2023.

To learn more about Infinite Athlete - http://infiniteathlete.ai/

Pro Football Sports Science Research Update

Does resisted sprint training improve the sprint performance of field-based invasion team sport players? A systematic review and meta-analysis. Sports Medicine, Ahead of Print: 1-14, 2023.

Does the operating temperature of accelerometers affect data measurement during static and dynamic conditions in sports? A randomized-control trial. Proceedings of the Institution of Mechanical Engineers, Part P: Journal of Sports Engineering and Technology, Ahead of Print: 1-7, 2023.

Effects of a strength and conditioning offseason program on countermovement jump ground reaction forces in Division 1 American football players. Journal of Strength and Conditioning Research, Ahead of Print: 1-10, 2023.

Effects of sprint versus strength training on risk factors for hamstring injury in football players. The Journal of Sports Medicine and Physical Fitness, 63(4): 580-587, 2023.

Effects of timing and types of protein supplementation on improving muscle mass, strength, and physical performance in adults undergoing resistance training: A network meta-analysis. International Journal of Sport Nutrition and Exercise Metabolism, 34: 54-64, 2024.

Epidemiology of stingers in the National Football League, 2015-2019. Sports Health, Ahead of Print: 1-8, 2024.

Heart rate variability in concussed college athletes: Follow-up study and biological sex differences. Brain Sciences, 13: 1669, 2023.

Here comes the STRAIN: Analyzing defensive pass rush in American football with player tracking data. The American Statistician, Ahead of Print: 1-10, 2023.

Hydration, hyperthermia, glycogen, and recovery: Crucial factors in exercise performance – A systematic review and meta-analysis. Nutrients, 15: 4442, 2023.

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

Pro Football Sports Science Research Update

Intermittent Fasting: Does it affect sports performance? A systematic review. Nutrients, 16: 168, 2024.

Investigating trade-offs made by American football linebackers using tracking data. Journal of Quantitative Analysis in Sports, 19(3): 171-185, 2023.

Jones fracture in the National Football League. Sports, 12: 7, 2024.

Mechanisms of mechanical overload-induced skeletal muscle hypertrophy: Current understanding and future directions. Physiological Reviews, 103: 2679-2757, 2023.

Mental health of elite sport coaches and entourage. Clinics in Sports Medicine, Ahead of Print: 1-13, 2023.

Monitoring the reconditioning of the injured football player with field-based measures: A case study following ACL reconstruction. ASPETAR Sports Medicine Journal, 12: 374-381, 2023.

Overview of the impact of sleep monitoring on optimal performance, immune system function and injury risk reduction in athletes: A narrative review. Science Progress, Vol. 106(4): 1-18, 2023.

Peak instantaneous PlayerLoad metrics highlight movement strategy deficits in professional male soccer players. Research in Sports Medicine, Ahead of Print: 1-12, 2022.

Possible relationship between the gut leaky syndrome and musculoskeletal injuries: The important role of gut microbiota as indirect modulator. AIMS Public Health, 10(3): 710-738, 2023.

Recovery of the autonomic nervous system following football training among Division 1 collegiate football athletes: The influence of intensity and time. Heliyon, e18125, 2023.

Risk of two sport-related concussions in the same year: Is the second concussion worse? Clinical Journal of Sport Medicine, 34: 38-43, 2024.

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

Pro Football Sports Science Research Update

Specific collagen peptides increase adaptations of patellar tendon morphology following 14-weeks of high-load resistance training: A randomized-controlled trial. European Journal of Sport Science, Vol. 23 (12): 2329-2339, 2023.

Sports injuries and game stakes: Concussions in the National Football League. Economic Inquiry, Ahead of Print: 1-19, 2023.

Strength and power thresholds to identify high and low linear sprint speed performers in collegiate American football players. Journal of Strength and Conditioning Research, Ahead of Print: 1-6, 2023.

The acute and chronic effects of implementing velocity loss thresholds during resistance training: A systematic review, meta-analysis and critical evaluation of the literature. Sports Medicine, 53(1): 177-214, 2023.

The ALR-RSI score can be used to evaluate psychological readiness to return to sport after acute Achilles tendon tear. Knee Surgery, Sports Traumatology, Arthroscopy. 31: 4961-4968, 2023.

The impact of eccentric or isometric training on strength, architecture, and sprinting across an Australian football season. Medicine and Science in Sports and Exercise, Ahead of Print: 1-45, 2023.

The multifaceted nature of recovery after exercise: A need for individualization. Sports Orthopaedics and Traumatology, Ahead of Print: 1-9, 2023.

Tracking devices and physical performance analysis in team sports: A comprehensive framework for research-trends and future directions. Frontiers in Sports and Active Living, 5:1284086, 2023.

Ultrasound shear wave seeds reduced following hamstring strain injury but not after returning to sport. Insights into Imaging, 15:7, 2024.

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

Pro Football Sports Science Research Update

Unlocking the healing potential: Cannabinoids in spine surgery for pain relief and recovery. JB&JS Reviews, 11(11):e23.00141, 2023.

Use of extracorporeal shockwave therapies for athletes and physically active individuals: A systematic review. British Journal of Sports Medicine, Ahead of Print: 1-12, 2024.

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

Pro Football Sports Science Research Update

A practical guide to improve sleep and performance in athletes. International Journal of Sports Science & Coaching, Ahead of Print: 1-12, 2023.

Abstract - Sleep is vital for optimal mental and physical health. For athletes, optimizing sleep is becoming a popular strategy to enhance athletic performance. Athletes often complain of sleep problems including insufficient sleep and insomnia symptoms and are also at a higher risk for sleep disordered breathing. Sleep disorders and insufficient sleep can contribute to excessive sleepiness, daytime dysfunction, and performance problems. In contrast, better sleep provides benefits for physical health and athletic performance. For athletes, multiple factors can contribute to insufficient sleep. Sport-specific fac tors include frequent travel across time zones, competition and training schedules, high training loads, and sleeping in an unfamiliar environment. Non-sport-related factors include work, social, and family commitments; attitudes and sleeping beliefs; individual characteristics, such as chronotype or preference for morning or evening; and lifestyle choices. Fortunately, there are strategies that can be implemented to improve sleep in athletes including (a) education and emphasis on the importance of sleep; (b) sleep screening; providing extra sleep opportunities like (c) banking sleep and (d) napping; improving sleep hygiene like (e) proper light exposure; (f) a good pre-sleep routine; (g) a conducive sleep environment; (h) a strategy for supplementation; (i) utilizing circadian timing adjustments; and (j) jet lag management. Increased recognition of the importance of sleep from sport professionals and screening for sleep disorders and disturbances will be key for future athlete health, well-being, and performance.

Ammonia Inhalants: Use, misuse, and role in sports performance. Sports Health, Ahead of Print: 1-5, 2023.

Context: Ammonia inhalants, also known as smelling salts, are preparations of ammonia designed to treat fainting but more commonly used by athletes to boost awareness and arousal during competition. Despite their widespread use, the physiological and performance-enhancing effects of ammonia inhalants remain poorly understood. The aim of the present study was to review the current literature surrounding the benefits, risks, and physiological effects of ammonia inhalants.

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

Pro Football Sports Science Research Update

Evidence Acquisition: An extensive literature review of articles pertaining to ammonia inhalants was performed through MEDLINE and Google Scholar. The search terms “smelling salts,” “ammonia inhalants,” “strength,” “performance,” “head injury,” and “concussion” were used.

Study Design: Clinical review.

Level of Evidence: Level 4.

Results: The physiological response to acute ammonia inhalation includes cerebral vasodilation and heart rate elevation without change in blood pressure. The existing evidence demonstrates an ergogenic benefit to ammonia inhalant use only during repeated bouts of high-intensity exercise; in these subjects, ammonia inhalation was associated with increased power as measured by the Wingate anaerobic test. In contrast, there is no performance benefit to ammonia inhalants in a short burst of maximal effort despite elevated arousal and an associated perception of performance enhancement. Importantly, ammonia inhalants have no role in medical management of head injuries, as they have the potential to exacerbate an underlying brain injury due to the involuntary withdrawal reflex associated with ammonia inhalation. Furthermore, the signs and symptoms of a concussion or more threatening head injury may be masked by ammonia inhalation and lead to continued participation in competition, causing additional harm.

Conclusion: Ammonia inhalants have no role in medical management of head injuries and have limited benefit with regards to sports performance.

Strength of Recommendation: B

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

Pro Football Sports Science Research Update

Are return-to-play times longer in lower-limb muscle injuries involving the intramuscular tendon? A systematic review. Journal of Science and Medicine in Sport. Ahead of Print: 1-11, 2023.

Objectives: To determine the influence of severity and direction (craniocaudal length vs cross-sectional area) of intramuscular tendon tears in the lower limb on return-to-play times, compared to muscle injuries without intramuscular tendon involvement.

Design: Systematic review with meta-analysis.

Methods: AMED, CINAHL, SPORTDiscus, ScienceDirect, PubMed (MEDLINE) and Web-of-Science were searched from inception to 31st July 2023, retrieving 666 records, of which nine were deemed eligible. A random-effects meta-analysis was performed on time to return-to-play for British Athletics Muscle Injury Classification 'b' vs 'c'.

Results: On the Quality in Prognosis Studies tool, one study had low risk of bias and eight had high risk. Using a best-evidence synthesis, no strong evidence emerged for a difference in time to return-to-play between injuries with or without intramuscular tendon involvement. Moderate evidence was found for an association between increased return-to-play time and the presence of "waviness" on magnetic resonance imaging and loss of tendon tension, but no association with longitudinal extent of tendon involvement. Pooled analysis revealed a medium effect-size difference between British Athletics Muscle Injury Classification 'b' and 'c' injuries, favoring classification 'b' (Hedges g = 0.67; 95% confidence interval 0.20 to 1.15; P = 0.002).

Conclusions: It remains difficult to provide an accurate prognosis for muscle injuries involving the intramuscular tendon due to high risk of bias and moderate heterogeneity across studies. Moderate evidence favored the prognosis for injuries at the musculotendinous junction (British Athletics Muscle Injury Classification 'b') over intratendinous injuries (British Athletics Muscle Injury Classification 'c').

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

Pro Football Sports Science Research Update

Associations between cognitive function and ACL injury-related biomechanics: A systematic review. Sports Health, Vol. 15(6): 855-866, 2023.

Context: Does lower baseline cognitive function predispose athletes to ACL injury risk, especially when performing unplanned or dual-task movements?

Objective: To evaluate the association between cognitive function and biomechanics related to ACL injuries during cognitively challenging sports movements.

Data Sources: PubMed (MEDLINE), Web of Science, Scopus, and SciELO databases were searched; additional hand searching was also conducted.

Study Selection: The following inclusion criteria had to be met: participants completed (1) a neurocognitive test, (2) a cognitively challenging sport-related task involving lower limbs, and (3) a biomechanical analysis. The following criteria determined exclusion from the review: studies involving participants with (1) recent or current musculoskeletal injuries; (2) recent or current concussion; (3) ACL surgical reconstruction, reviews of the literature, commentary or opinion articles, and case studies.

Study Design: Systematic review using the Preferred Reporting Items for Systematic Reviews and Meta-Analysis Protocols (PRISMA-P) statement and registered at the International Prospective Register of Systematic Reviews (PROSPERO).

Level of Evidence: Level 3.

Data Extraction: Two of authors independently extracted data and assessed the methodological quality of the articles with the Downs and Black and ROBINS-I checklists, to assess methodological quality and risk of bias, respectively.

Results: Six studies with different methodologies and confounding factors were included in this review. Of these 6 studies, 3 were ranked as high-quality, 3 demonstrated a low risk of bias, 2 a moderate risk, and 1 a severe risk. Five studies found a cognitive-motor relationship, with worse cognitive performance associated with increased injury risk, with 1 study reporting the opposite directionality for 1 variable. One study did not identify any interaction between cognitive function and biomechanical outcomes.

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

Pro Football Sports Science Research Update

Conclusion: Worse cognitive performance is associated with an increased injury risk profile during cognitively challenging movements.

Athletes can benefit from increased intake of EPA and DHA – Evaluating the evidence. Nutrients, 15: 4925, 2023.

Abstract - Fatty fish, which include mackerel, herring, salmon and sardines, and certain species of algae (e.g., Schizochytrium sp., Crytthecodiniumcohnii and Phaeodactylumtricornutum) are the only naturally rich sources of the omega-3 polyunsaturated fatty acids (n-3 PUFAs) eicosapentaenoic acid (EPA) and docosahexaenoic acid (DHA). EPA and DHA are the most biologically active members of the n-3 PUFA family. Limited dietary sources and fluctuating content of EPA and DHA in fish raise concerns about the status of EPA and DHA among athletes, as confirmed in a number of studies. The beneficial effects of EPA and DHA include controlling inflammation, supporting nervous system function, maintaining muscle mass after injury and improving training adaptation. Due to their inadequate intake and beneficial health-promoting effects, athletes might wish to consider using supplements that provide EPA and DHA. Here, we provide an overview of the effects of EPA and DHA that are relevant to athletes and discuss the pros and cons of supplements as a source of EPA and DHA for athletes.

Blood and MRI biomarkers of mild traumatic brain injury in non-concussed collegiate football players. Scientific Reports, 14: 665, 2024.

Abstract - Football has one of the highest incidence rates of mild traumatic brain injury (mTBI) among contact sports; however, the effects of repeated sub-concussive head impacts on brain structure and function remain under-studied. We assessed the association between biomarkers of mTBI and structural and functional MRI scans over an entire season among non-concussed NCAA Division I linemen and non-linemen. Concentrations of S100B, GFAP, BDNF, NFL, and NSE were assessed in 48 collegiate football players (32 linemen; 16 non-linemen) before the start of pre-season training (pre-camp), at the end of pre-season training (pre-season), and at the end of the competitive season (post-season). Changes in brain structure and function were assessed in a sub-sample of 11 linemen and 6 non-linemen using

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

structural and functional MRI during the execution of Stroop and attention network tasks. S100B, GFAP and BDNF concentrations were increased at post-season compared to pre-camp in linemen. White matter hyperintensities increased in linemen during pre-season camp training compared to pre camp. This study showed that the effects of repeated head impacts are detectable in the blood of elite level non-concussed collegiate football players exposed to low-moderate impacts to the heads, which correlated with some neurological outcomes without translating to clinically-relevant changes in brain anatomy or function.

Changes in health-related parameters associated with sports performance drugs. International Journal of Sports Medicine, 44(3): 206-214, 2023.

Abstract - The purpose of this study was to evaluate changes in health-related parameters caused by the administration of anabolic androgenic steroids and "fat-burning drugs" during a 6-month competition preparation period. The physiological, biochemical, and anthropometric parameters studied included serum cholesterol, triglycerides, high-density lipoprotein cholesterol, low-density lipoprotein cholesterol, aspartate aminotransferase, alanine transaminase, bilirubin, body mass, and per centage of total body fat. Changes in the parameters studied were analyzed at monthly intervals during six months of preparation for competition. The study revealed a continuous increase in body mass, accompanied by a decrease in body fat percentage to the physiologically essential level. Total cholesterol levels remined in the desirable concentration range. The mean levels of triglycerides fluctuated between borderline high and high. Mean high-density lipoprotein cholesterol levels remained within the low range, while low-density lipoprotein cholesterol fluctuated between near-optimal / above-optimal, borderline high, and high levels. Serum levels of aspartate aminotransferase and alanine transaminase remained within the high concentration. The bilirubin concentration remained in the desirable range. The blood nitrogen urea concentration fluctuated between normal and elevated levels. Sports-enhancing drugs analyzed in this study do not have an immediate detrimental impact on the selected biochemical, physiological, and anthropometric parameters that define health.

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

Pro Football Sports Science Research Update

Decreased concussion incidence following the implementation of the Targeting Rules: An updated epidemiology of National Football League concussions from 2017 to 2022. Cureus, 15(12): e50997, 2023.

Abstract - The incidence of concussions in football, and the ensuing media attention, has garnered scientific investigation, prompted technological advances in protective gear, and altered the rules of the game, including the National Football League's (NFL) "Targeting" rule, which began in 2018, but the impact of these changes is unclear. This study aims to describe the epidemiology of concussions that occurred in five NFL seasons from the 2017-2018 season through the 2021-2022 season and characterize positional differences in rate and games missed. There was a significant decrease ($p = 0.02$) in total concussions between the 2017-2018 season (102 concussions) and the remaining four seasons (average of 73.80 concussions per year), accounting for a 38% decrease. Offensive and defensive units had decreased concussion rates and average games missed per concussion. Defensive backs (10.46 per 1,000 athlete exposures (AEs)) and tight ends (10.69 per 1,000 AEs) had the highest concussion rates, and the defensive line had the highest average games missed per concussion at 3.97. The introduction of the "Targeting" rule and other rule changes in the NFL in 2018 correlated with a decrease in total concussions per year, total games missed due to concussion, and average games missed per concussion. Offense and defense experienced similar reductions in concussion incidence and severity. Overall, the updated epidemiology of NFL concussions suggests that the incidence of concussions has decreased; however, players continue to experience concussions that require them to miss multiple games.

Does increased core temperature alter cognitive performance during exercise-induced heat strain? A narrative review. 135: 35-52, 2023.

Abstract - Research to date provides equivocal evidence regarding the influence of heat stress, heat strain, and more specifically, elevated exercise-induced core temperature on cognitive performance. This review sought to identify differences in how specific cognitive tasks were affected by increases in core body temperatures. Included papers (n = 31) measured cognitive performance and core temperature during exercise, while experiencing heightened thermal stress. Cognitive tasks were classified as cognitive inhibition, working memory, or cognitive flexibility tasks. Independently, core

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

temperature changes were not sufficient predictors of cognitive performance. However, reaction time, memory recall, and Stroop tasks appeared to be most effective at identifying cognitive changes during heightened thermal strain. Alterations in performance were more likely to arise under increased thermal loads, which were typically associated with cumulative physiological stressors, such as elevated core temperatures, occurring alongside dehydration, and prolonged exercise durations. Future experimental designs should consider the relevance, or futility of assessing cognitive performance in activities that do not elicit a considerable degree of heat strain, or physiological load.

Does resisted sprint training improve the sprint performance of field-based invasion team sport players? A systematic review and meta-analysis. Sports Medicine, Ahead of Print: 1-14, 2023.

Background: Developing the sprint performance of field-based invasion team sport (FITS) players is considered an essential training goal for FITS coaching practitioners, and thus numerous training methods are employed to elicit improvements. Although interest in resisted sprint training (RST) has grown considerably in recent times, there remains a lack of clarity around its utility in FITS, particularly regarding the use and effectiveness of heavier RST loads.

Objectives: The aims of this review were to (1) compare RST to unresisted sprinting, (2) examine if RST can improve sprint performance and (3) investigate if external load and the method of load prescription influence the impact of RST in FITS players.

Methods: The systematic review and meta-analysis were conducted in compliance with the Preferred Reporting Items for Systematic Review and Meta-Analyses (PRISMA) guidelines. The search strategy included terms for RST, RST modalities and FITS, and was applied to PubMed, SPORTDiscus, Web of Science and OpenGrey databases. Methodological quality and risk of bias associated with each study were assessed using the Physiotherapy Evidence Database scale (PEDro) and Cochrane Risk of Bias assessment tool respectively.

Results: Twenty-one studies met the inclusion criteria for this review and were included in the final analysis. The primary between-group analysis revealed no differences between RST and unresisted

sprinting for developing the early acceleration, late acceleration and maximum velocity sprint phases. Secondly, a within-group analysis found significant improvements for resisted sprint training in the early acceleration (standardized mean difference [SMD] −0.80) and late acceleration (SMD−0.28) sprint phases, with no change detected for the maximum velocity phase. Finally, significant moderate improvements were found for light (SMD−0.69) and very heavy (SMD−1.01) loads during early acceleration.

Conclusions: Resisted sprint training achieved similar improvements in sprint performance to those found for unresisted sprinting during the acceleration and maximum velocity sprint phases. Within-group findings show RST is an effective method for improving early acceleration and late acceleration performance of FITS players. Finally, a subgroup analysis supports the use of light and very heavy loads for increasing early acceleration performance, while also highlighting greater benefits associated with using the percentage velocity decrement loading method.

Key Points

- No differences were found between resisted sprint training and unresisted sprinting for increasing the sprint performance of field-based invasion team sport players.
- Resisted sprint training is an effective method to increase the acceleration performance of field-based invasion team sport players with early acceleration demonstrating a larger effect size (standardized mean diference−0.80) than late acceleration (standardized mean diference−0.28).
- Light (2.5–10% velocity decrement) and very heavy (≥50% velocity decrement) loads were found effective for improving early acceleration performance. This result may suggest that performing resisted sprint training with loads spanning large ranges of the percentage velocity decrement loading method will invariably elicit improvements to early acceleration in comparison to the late acceleration and maximum velocity phases where there may be a requirement for greater velocity specificity.

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

Pro Football Sports Science Research Update

Does the operating temperature of accelerometers affect data measurement during static and dynamic conditions in sports? A randomized-control trial. Proceedings of the Institution of Mechanical Engineers, Part P: Journal of Sports Engineering and Technology, Ahead of Print: 1-7, 2023.

Abstract - This study aimed to analyze the effect of two operating temperatures (optimal vs environmental) on IMU internal temperature and accelerometry-based data in two conditions of measurement: (a) static, no movement; and (b) dynamic, a U'16 game-based training soccer session with 14 players. Each session took a 60-min duration and followed an ecological design (no intervention in environmental conditions). Fifty-six WIMU PROTM inertial devices were randomly grouped in four conditions (two operating temperatures 3 two measurement conditions, 14 devices in each group) and recorded IMUs' internal temperature, resultant acceleration vector and PlayerLoad (PLRT). Devices were placed in the manufacturer's docking station at the static condition and in a specific harness at the interscapular level at the dynamic condition. Repeated measures t-tests were employed for between-group comparisons at 1-min average sections throughout the register. Between-group differences were found in the static and dynamic conditions in IMUs internal temperature (static: 0–31 min; dynamic: 0–20 min) and resultant acceleration vector (static: 0–29 min; dynamic: 0–19 min). Instead, no differences were found in PLRT dynamics and accumulated PLRT although values in the environmental group were overestimated. In conclusion, WIMU PROTM devices must achieve the optimal operating temperature (38–39^0) to avoid measurement error, regardless of the application (short or long sessions).

Effects of a strength and conditioning offseason program on countermovement jump ground reaction forces in Division 1 American football players. Journal of Strength and Conditioning Research, Ahead of Print: 1-10, 2023.

Abstract - The purpose of this study was to examine the effects of a 10-week strength and conditioning offseason program on the ground reaction forces (GRFs) of American football players during single-leg and double-leg countermovement jumps (SLJ and CMJ, respectively). Each subject visited the laboratory twice, once for pre-offseason and once for post-offseason testing. During each visit, subjects performed

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

CMJs and SLJs for each leg. Ground reaction forces were collected by force plates to quantify unweighting, braking, propulsive, and performance metrics for each jump. In addition, an efficiency index was calculated for each jump to examine changes in vertical jump efficiency. Dependent samples t tests compared all CMJ metrics. Two-way repeated measures analyses of variance (leg × time) compared all SLJ metrics. An alpha level of $p \leq 0.05$ was considered statistically significant. For the CMJ, propulsive phase duration decreased due to the program ($p = 0.007$), whereas peak braking power, peak propulsive power, mean propulsive force, and jump height increased ($p \leq 0.012$). For the SLJ, peak braking power, force at the low position, braking rate of force development, eccentric force, peak propulsive power, mean propulsive force, and jump height increased in both legs ($p \leq 0.044$). The efficiency index increased for the CMJ and the SLJ for both legs ($p \leq 0.016$). This study demonstrated that SLJ and CMJ vertical jump performance significantly increases in as few as 10 weeks of offseason strength and conditioning. Strength and conditioning programming may effectively increase vertical jump performance, as assessed by GRFs, which can be used as a simple indicator regarding changes in athletic performance.

Practical Applications - Although not surprising that vertical jump performance increased after an offseason strength and conditioning program, having a greater understanding of the underlying physiological adaptations of muscle can help strength and conditioning professionals maximize their training programs. This study demonstrates that an offseason training program of 10 weeks may effectively enhance eccentric and concentric capabilities of muscle. This provides further insight into which metrics may be important to track. The ability to absorb and reapply forces during the eccentric phase of muscular contraction is a vital component of sprinting, change of direction, and agility tasks, all of which occur in American football. In addition, unweighting and braking phase metrics may allow practitioners to identify key factors leading to “energy leakage” during the eccentric phase of a jump. This in turn allows practical application of this knowledge into programming for each athlete, aiding in creating individualized programs for each athlete. Furthermore, GRF metrics also allow practitioners to identify efficiency patterns in athletes, which can lead to greater accuracy regarding force, velocity, and power, capabilities of their athletes. This may in turn allow for more individualized programming using velocity-based training principles, which have grown in popularity within the field of strength and conditioning. In addition, because the SLJ has been suggested as a possible indicator of injury risk, assessments of GRFs throughout a strength and conditioning program may not only provide insight regarding changes in performance but potentially may shed light on future injury risk. Thus, strength

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

and conditioning programming may effectively increase vertical jump performance, as assessed by GRFs, which can be used as a simple indicator regarding changes in athletic performance. Furthermore, coaches and practitioners may find value in regular assessments of ground reaction forces. In conjunction with the growing field of sport science in athletic programs, vertical jump performance data may be more effectively monitored and used to modify and adapt current strength and conditioning programs based on the results of regular GRF assessments.

Effects of sprint versus strength training on risk factors for hamstring injury in football players. The Journal of Sports Medicine and Physical Fitness, 63(4): 580-587, 2023.

Background: This study aimed to compare the effects of in-season sprint training vs. Nordic hamstring exercise (NHE) training on risk factors for hamstring strain injuries (HSI).

Methods: Eighteen male university football players (20.9±2.5 years; 181±7 cm; 75.8±9.1 kg; 15.2±3.5% of body fat) were randomly allocated to a sprint group or NHE group. They completed baseline isokinetic strength and sprint mechanics assessments prior to their assigned intervention performed twice weekly for 4-weeks, before post-testing. A mixed design ANOVA with repeated measures assessed time, group and interaction effects for all risk factors.

Results: There were significant increases in hamstring eccentric peak torque at 60°·s-1 (+8% - 9.9%), the torque produced at 20° (+15%) and 10° (+21% - 31%), as well as a rightward shift in angle of peak torque towards knee extension (-27% - -36%) in both groups ($P<0.05$). We also observed a significant increase (+24.5%) in hamstring eccentric peak torque at 180°·s-1 in the strength group only and significant improvements (+29.4%) in the rate of torque development of the dominant leg at 60°·s-1 in the sprint group only ($P<0.05$). No significant effect was noted on sprint performance or sprint mechanics ($P>0.05$).

Conclusions: These findings suggest that both training programs can be effective to mitigate the risk of HSI, but through different mechanisms.

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

Pro Football Sports Science Research Update

Effects of timing and types of protein supplementation on improving muscle mass, strength, and physical performance in adults undergoing resistance training: A network meta-analysis. International Journal of Sport Nutrition and Exercise Metabolism, 34: 54-64, 2024.

Abstract - Precise protein supplementation strategies for muscle improvement are still lacking. The timing or type of protein supplementation has been debated as a window of opportunity to improve muscle mass, strength, and physical performance. We conducted a network meta-analysis of randomized controlled trials with protein supplements and resistance training. PubMed, Web of Science, Cochrane Library, and SPORTDiscus databases were searched until May 1, 2023. We included 116 eligible trials with 4,711 participants that reported on 11 timing and 14 types of protein supplementation. Compared with placebo, protein supplementation after exercise (mean difference [MD]: 0.54 kg [95% confidence intervals 0.10, 0.99] for fat-free mass, MD: 0.34 kg [95% confidence intervals 0.10, 0.58] for skeletal muscle mass) and at night (MD: 2.85 kg [0.49, 5.22] for handgrip strength, MD: 12.12 kg [3.26, 20.99] for leg press strength) was most effective in improving muscle mass and strength, respectively (moderate certainty). Milk proteins (milk, whey protein, yogurt, casein, and bovine colostrum), red meat, and mixed protein were effective for gains in both muscle mass and strength (moderate certainty). No timing or type of protein showed a significant enhancement in physical performance (timed up-to-go test, 6-min walk test, and gait speed). Pre/postexercise and Night are key recommended times of protein intake to increase muscle mass and strength, respectively. Milk proteins are the preferred types of protein supplements for improving muscle mass and strength. Future randomized controlled trials that directly compare the effects of protein timing or types are needed.

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

Pro Football Sports Science Research Update

Epidemiology of stingers in the National Football League, 2015-2019. Sports Health, Ahead of Print: 1-8, 2024.

Background: Transient traumatic neuropraxia of either the brachial plexus or cervical nerve root(s) is commonly described as a "stinger" or "burner" by the athlete. Stingers in American Football commonly occur acutely as isolated injuries; however, concomitant injuries, including cervical spine pathologies, have also been reported.

Hypothesis: Among National Football League (NFL) athletes, the incidence rate of stingers is higher during the regular season than during the preseason and among positions with high velocity impacts such as running backs, linebackers, defensive backs, and receivers.

Study Design: Retrospective epidemiology study.

Level of Evidence: Level 4.

Methods: Aggregation of all in-game injuries with a clinical impression of "neck brachial plexus stretch" or "neck brachial plexus compression" entered into the NFL injury surveillance database through the centralized league-wide electronic medical record system over 5 years (2015-2019 seasons). Incidence rates per player-play were calculated and reported.

Results: A total of 691 in-game stingers occurred during the study period, with a mean of 138.2 per year. Average single season injury risk for incident stinger was 3.74% (95% CI, 3.46%-4.05%). The incidence rate was higher during regular season games than during preseason games (12.26 per 100,000 player-plays [11.30-13.31] vs 8.87 [7.31-10.76], $P < 0.01$, respectively). The highest reported stinger incidence rates were among running backs and linebackers (both >15 per 100,000 player-plays). Among stingers, 76.41% did not miss time. Of those that resulted in time lost from football activities, mean time missed due to injury was 4.79 days (range, 3.17-6.41 days). Concomitant injuries were relatively low (7.09%).

Conclusion: In-game stinger incidence was stable across the study period and occurred most frequently in running backs and linebackers. Stingers were more common during the regular season, and most players did not miss time. Concomitant injuries were relatively rare.

Clinical Relevance: An improved understanding of the expected time loss due to stinger and concomitant injuries may provide insight for medical personnel in managing these injuries.

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

Pro Football Sports Science Research Update

Heart rate variability in concussed college athletes: Follow-up study and biological sex differences. Brain Sciences, 13: 1669, 2023.

Abstract - Finding reliable biomarkers to assess concussions could play a pivotal role in diagnosis, monitoring, and predicting associated risks. The present study aimed to explore the use of heart rate variability (HRV) in the follow-up of concussions among college athletes and to investigate the relationships between biological sex, symptomatology, and HRV values at baseline and after a concussion. Correlations between measures were also analyzed. A total of 169 (55 females) athletes aged 16 to 22 years old completed baseline testing, and 30 (8 females) concussion cases were followed. Baseline assessment (T1) included psychosocial and psychological questionnaires, symptoms report, and four minutes of HRV recording. In the event of a concussion, athletes underwent re-testing within 72 h (T2) and before returning to play (T3). Baseline findings revealed that girls had higher %VLF while sitting than boys, and a small negligible correlation was identified between %HF and total symptoms score as well as %HF and affective sx. Post-concussion analyses demonstrated a significant effect of time × position × biological sex for %HF, where girls exhibited higher %HF at T3. These findings suggest disruptions in HRV following a concussion and underscore biological sex as an important factor in the analysis of HRV variation in concussion recovery trajectory.

Here comes the STRAIN: Analyzing defensive pass rush in American football with player tracking data. The American Statistician, Ahead of Print: 1-10, 2023.

Abstract - In American football, a pass rush is an attempt by the defensive team to disrupt the offense and prevent the quarterback (QB) from completing a pass. Existing metrics for assessing pass rush performance are either discrete-time quantities or based on subjective judgment. Using player tracking data, we propose STRAIN, a novel metric for evaluating pass rushers in the National Football League (NFL) at the continuous-time within-play level. Inspired by the concept of strain rate in materials science, STRAIN is a simple and interpretable means for measuring defensive pressure in football. It is a directly observed statistic as a function of two features: the distance between the pass rusher and QB, and the rate at which this distance is being reduced. Our metric possesses great predictability of pressure and stability over time. We also fit a multilevel model for STRAIN to understand the defensive

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

pressure contribution of every pass rusher at the play-level. We apply our approach to NFL data and present results for the first eight weeks of the 2021 regular season. In particular, we provide comparisons of STRAIN for different defensive positions and play outcomes, and rankings of the NFL's best pass rushers according to our metric.

Hydration, hyperthermia, glycogen, and recovery: Crucial factors in exercise performance – A systematic review and meta-analysis. Nutrients, 15: 4442, 2023.

Abstract - Hyperthermia accelerates dehydration and can lead to a glycolysis malfunction. Therefore, to deeply understand the relationship between dehydration and hyperthermia during exercise, as well as in the recovery time, there might be important factors to improve athletic performance. A systematic review was carried out in different databases using the words "hydration" OR "dehydration" AND "glycogen" OR "glycogenesis" OR "glycogenolysis" AND "muscle" OR "muscle metabolism" OR "cardiovascular system" and adding them to the "topic section" in Web of Science, to "Title/Abstract" in PubMed and to "Abstract" in SPORTDiscus. A total of 18 studies were included in the review and 13 in the meta-analysis. The free statistical software Jamovi was used to run the meta-analysis (version 1.6.15). A total sample of 158 people was included in the qualitative analysis, with a mean age of 23.5 years. Ten studies compared muscle glycogen content after hydration vs. remaining dehydrated (SMD −4.77 to 3.71, positive 80% of estimates, \hat{\mu} = 0.79 (95% CI: −0.54 to 2.12), z = 1.17, p = 0.24, Q-test (Q(9) = 66.38, p < 0.0001, tau^2 = 4.14, I 2 = 91.88%). Four studies examined the effect of temperature on postexercise muscle glycogen content (SMD −3.14 to −0.63, 100% of estimates being negative, \hat{\mu} = −1.52 (95% CI: −2.52 to −0.53), (z = −3.00, p = 0.003, Q-test (Q(3) = 8.40, p = 0.038, tau^2 = 0.68, I 2 = 66.81%). In conclusion, both hyperthermia and dehydration may contribute to elevated glycogenolysis during exercise and poor glycogen resynthesis during recovery. Although core and muscle hyperthermia are the key factors in glycogen impairments, they are also directly related to dehydration.

Pro Football Sports Science Research Update

Intermittent Fasting: Does it affect sports performance? A systematic review. Nutrients, 16: 168, 2024.

Abstract - Intermittent fasting is one of the most popular types of diet at the moment because it is an effective nutritional strategy in terms of weight loss. The main objective of this review is to analyze the effects that intermittent fasting has on sports performance. We analyzed physical capacities: aerobic capacity, anaerobic capacity, strength, and power, as well as their effect on body composition. For this, a bibliographic search was carried out in several databases where 25 research articles were analyzed to clarify these objectives. Inclusion criteria: dates between 2013 and present, free full texts, studies conducted in adult human athletes, English and/or Spanish languages, and if it has been considered that intermittent fasting is mainly linked to sports practice and that this obtains a result in terms of performance or physical capacities. This review was registered in PROSPERO with code ref. 407024, and an evaluation of the quality or risk of bias was performed. After this analysis, results were obtained regarding the improvement of body composition and the maintenance of muscle mass. An influence of intermittent fasting on sports performance and body composition is observed. It can be concluded that intermittent fasting provides benefits in terms of body composition without reducing physical performance, maintenance of lean mass, and improvements in maximum power. But despite this, it is necessary to carry out new studies focusing on the sports field since the samples have been very varied. Additionally, the difference in hours of intermittent fasting should be studied, especially in the case of overnight fasting.

Investigating trade-offs made by American football linebackers using tracking data. Journal of Quantitative Analysis in Sports, 19(3): 171-185, 2023.

Abstract - In recent years, the game of football has made a shift towards being more quantitative. With the advent of charting and tracking data, player evaluation is able to be studied from several different angles. In this paper, we build and refine two novel metrics: Bite Distance Under Expected (BDUE) and Ground Covered Over Expected (GCOE) for the evaluation of linebackers in the National Football League (NFL). Here, we show that these metrics are heavily correlated with each other, which demonstrates the trade-off linebackers have to make between being aggressive against the run and being effective when

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

the opposing offense is using play-action. We also show that these metrics are more stable than those in the public space. Finally, we show how these metrics measure deception by opposing offenses.

Jones fracture in the National Football League. Sports, 12: 7, 2024.

Background: Jones fracture, or proximal fifth metatarsal fracture, is a common injury in National Football League (NFL)-caliber athletes. Combine draft performance can greatly impact the long-term success of these athletes, and substantial emphasis has been placed on early return to play (RTP) and the minimization of post-operative complications after Jones fracture in these athletes. To date, no study has specifically described the treatment and outcomes of this injury specifically in NFL-caliber players, considering factors relevant to this unique population. Thus, the purpose of this review is to delve into Jones fracture in NFL-caliber athletes, evaluating the diagnostic, treatment, and RTP considerations.

Methods: We searched Medline (PubMed), Embase (Elsevier), Scopus (Elsevier), and SPORTDiscus (EBSCOhost) for the concept of Jones fractures in the NFL. Using the PRISMA guidelines, a team of three reviewers conducted abstract screenings, full-text screenings, and the extraction of studies describing Jones fractures specifically in the NFL.

Results: Of the 1911 studies identified, 6 primary retrospective studies met the inclusion and exclusion criteria. The heterogeneity of the outcome reporting precluded a meta-analysis; thus, a qualitative review of manuscripts describing Jones fracture was carried out. Classification, diagnosis, and treatment considerations, RTP statistics and outcomes, and complications were discussed. Amongst the primary studies, there were 285 Jones fractures, all athletes were able to RTP, and the average time to RTP ranged from 6 weeks to 27 weeks. For complications, with operative treatment, the refracture rate ranged from 4 to 12%, and incomplete healing ranged from 7 to 50%. RTP was 15 weeks for refractures. There were no patient-reported outcomes.

Conclusions: The vast majority of Zone 2/3 Jones fractures are treated with IM screw fixation with or without adjunctive orthobiologics, such as bone marrow aspirate concentrate, in NFL-caliber athletes. The six major series investigating outcomes after the operative treatment of Jones fractures in NFL players reveal very positive findings overall with regard to RTP, reoperation, and career continuation.

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

Pro Football Sports Science Research Update

Mechanisms of mechanical overload-induced skeletal muscle hypertrophy: Current understanding and future directions. Physiological Reviews, 103: 2679-2757, 2023.

Abstract - Mechanisms underlying mechanical overload-induced skeletal muscle hypertrophy have been extensively researched since the landmark report by Morpurgo (1897) of "work-induced hypertrophy" in dogs that were treadmill trained. Much of the preclinical rodent and human resistance training research to date supports that involved mechanisms include enhanced mammalian/mechanistic target of rapamycin complex 1 (mTORC1) signaling, an expansion in translational capacity through ribosome biogenesis, increased satellite cell abundance and myonuclear accretion, and postexercise elevations in muscle protein synthesis rates. However, several lines of past and emerging evidence suggest that additional mechanisms that feed into or are independent of these processes are also involved. This review first provides a historical account of how mechanistic research into skeletal muscle hypertrophy has progressed. A comprehensive list of mechanisms associated with skeletal muscle hypertrophy is then outlined, and areas of disagreement involving these mechanisms are presented. Finally, future research directions involving many of the discussed mechanisms are proposed.

CLINICAL HIGHLIGHTS

- Loss of muscle mass with aging and certain noncommunicable diseases (e.g., cancer, COPD, and others) is associated with increased mortality. Thus, understanding the mechanisms controlling skeletal muscle hypertrophy can help determine the most effective interventions to preserve or enhance muscle mass.
- Studies in animals and humans suggest that mechanical overload (e.g., resistance training) best achieves skeletal muscle hypertrophy. Bouts of mechanical overload induce transient increases in mammalian/mechanistic target of rapamycin complex 1 (mTORC1) signaling leading to elevations in muscle protein synthesis rates. With repeated bouts of mechanical overload, these events contribute to skeletal muscle hypertrophy.
- An expansion in translational capacity through ribosome biogenesis and increases in satellite cell abundance and myonuclear accretion also contribute to skeletal muscle hypertrophy following days to weeks of repeated mechanical overload bouts.

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

Pro Football Sports Science Research Update

- Aside from these three aforementioned mechanisms, several lines of past, current, and emerging research suggest that other mechanisms may also contribute to mechanical overload-induced skeletal muscle hypertrophy (e.g., mTORC1-independent signaling, microRNAs, genetic polymorphisms, and enhanced angiogenesis among others).
- There are also potential manners in which epigenetic alterations in myonuclear and mitochondrial DNA, extracellular matrix remodeling, cytoskeletal remodeling, mitochondrial biogenesis, bioenergetic adaptations, and other mechanisms can contribute to mechanical overload-induced skeletal muscle hypertrophy.
- The current and rapidly emerging molecular tools available to researchers as well as rodent and human studies being performed in tandem will continue to provide insight into novel mechanisms that are needed for mechanical overload-induced skeletal muscle hypertrophy to occur.

Mental health of elite sport coaches and entourage. Clinics in Sports Medicine, Ahead of Print: 1-13, 2023.

KEY POINTS

- The mental health of elite coaches and entourage staff has been understudied, particularly in comparison to the mental health of elite athletes.
- Elite coaching and entourage staff are subject to similar sport-related stressors and mental health risk factors as elite athletes, such as performance demands, scrutiny, and travel away from home for sport.
- Available evidence, while scarce, shows that elite coaches and entourage experience depression, anxiety, and psychological distress at similar levels to elite athlete samples.
- Additional research is needed to strengthen understanding of the extent of mental ill-health and mental health needs among individuals working in elite-level sport alongside athletes.
- Based on the extant literature, mental health screening and interventions for elite athletes should be extended to the coaches and entourage staff working in the same high-performance environments.

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

Pro Football Sports Science Research Update

Monitoring the reconditioning of the injured football player with field-based measures: A case study following ACL reconstruction. ASPETAR Sports Medicine Journal, 12: 374-381, 2023.

Overview - Current research suggests that ACL reconstructions should follow a criterion-based process over successive rehabilitation phases. Today there is increasing information on the progression criteria and methods to rehabilitate the neuromuscular system (i.e., range of motion (ROM), strength, balance, and control) and restore generic and football-specific functional capacities (e.g., jumping, landing, cutting, accelerating, and decelerating) across each of these phases. There is, however, less information and research on the metabolic conditioning aspect of the post-ACL rehabilitation process.

Mitigating cardiorespiratory fitness deconditioning or even developing an athlete's physical fitness to a higher level than before injury during the on-field rehabilitation phase may be an overlooked factor of a successful return-to-sport process. In fact, higher fitness levels translate into a lower relative internal load for a given workout, which puts players under optimal conditions when returning to train with the team (return to train phase that follows the on-field rehabilitation phase). This may support better decision-making and decrease cognitive load, which may, in turn, facilitate players' reintroduction to the intense stimuli of football practice. Importantly in this context, this also likely contributes in reducing the risk of re-injury.

Lastly, while there is a growing use of technology to monitor function and muscle strength in the gym (e.g., force plates, dynamometers), methods and tools that can be used in the field during the on-field rehabilitation are still limited.

In the present manuscript, we use the case of a recent ACL injury that happened to a Lille OSC player to introduce our post-ACL reconditioning approach during the first rehabilitation and on-field rehabilitation phases. We provide examples of metabolic work programming through various generic and football-specific drills. We also touch on field-based tools and strategies for monitoring fitness and running mechanics that were used to further guide the programming throughout the process.

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

Pro Football Sports Science Research Update

Overview of the impact of sleep monitoring on optimal performance, immune system function and injury risk reduction in athletes: A narrative review. Science Progress, Vol. 106(4): 1-18, 2023.

Abstract - Sleep is essential for a range of physiological and mental functions in professional athletes. There is proof that athletes may experience lower quality and quantity of sleep. While adequate sleep has been shown to have restorative effects on the immune system and endocrine system, facilitate nervous system recovery and the metabolic cost of wakefulness, and play a significant role in learning, memory, and synaptic plasticity, which can affect sports recovery, injury risk reduction, and performance. Sports performance may suffer significantly from a lack of sleep, especially under maximal and long-term exercise. Due to the potential harm, these factors may do to an athlete's endocrine, metabolic, and nutritional health, sports performance is impacted by reduced sleep quality or quantity. There are several neurotransmitters associated with the sleep–wake cycle that have been discovered. They comprise cholinergic hormone, orexin, melanin, galanin, serotonin, gamma-aminobutyric acid, histamine, and serotonin. Therefore, dietary modifications that affect the neurotransmitters in the brain also may affect sleep; particularly for athletes who require more physical and psychological recovery owing to the tremendous physiological and psychological demands placed on them during training and performance. This review explores the variables that influence the quantity and quality of sleep-in populations of athletes and assesses their possible effects. In addition, several recommendations for improving sleep are presented. Even though there has been much research on variables that impact sleep, future studies may high light the significance of these aspects for athletes.

Peak instantaneous PlayerLoad metrics highlight movement strategy deficits in professional male soccer players. Research in Sports Medicine, Ahead of Print: 1-12, 2022.

Abstract - To investigate the influence of task, limb dominance and previous injury on single leg hop task performance and loading response, 25 professional male soccer players completed anterior, medial and lateral hop tests with an accelerometer at mid-calf. Performance outcome was defined as hop distance with loading response defined as the magnitude of, and time to peak instantaneous planar PlayerLoad.

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

Pro Football Sports Science Research Update

The performance was sensitive to task and previous injury ($P < 0.001$) but not limb dominance, with no evidence of bilateral asymmetry ($P = 0.668$). Despite impaired performance, previously injured players did not exhibit lower peak instantaneous PlayerLoad after impact in any plane ($P \geq 0.110$). There was however a significantly ($P = 0.001$) longer time to peak medio-lateral loading after impact in previously injured players' affected limb. This observation was exacerbated when the injury was to the non-dominant limb ($P = 0.041$). Lower-limb accelerometry enhances understanding of movement strategy beyond task outcome, with practical implications in player screening and objective rehabilitation.

Possible relationship between the gut leaky syndrome and musculoskeletal injuries: The important role of gut microbiota as indirect modulator. AIMS Public Health, 10(3): 710-738, 2023.

Abstract - This article aims to examine the evidence on the relationship between gut microbiota (GM), leaky gut syndrome and musculoskeletal injuries. Musculoskeletal injuries can significantly impair athletic performance, overall health, and quality of life. Emerging evidence suggests that the state of the gut microbiota and the functional intestinal permeability may contribute to injury recovery. Since 2007, a growing field of research has supported the idea that GM exerts an essential role maintaining intestinal homeostasis and organic and systemic health. Leaky gut syndrome is an acquired condition where the intestinal permeability is impaired, and different bacteria and/or toxins enter in the bloodstream, thereby promoting systemic endotoxemia and chronic low-grade inflammation. This systemic condition could indirectly contribute to increased local musculoskeletal inflammation and chronificate injuries and pain, thereby reducing recovery-time and limiting sport performance. Different strategies, including a healthy diet and the intake of pre/probiotics, may contribute to improving and/or restoring gut health, thereby modulating both systemically as local inflammation and pain. Here, we sought to identify critical factors and potential strategies that could positively improve gut microbiota and intestinal health, and reduce the risk of musculoskeletal injuries and its recovery-time and pain. In conclusion, recent evidences indicate that improving gut health has indirect consequences on the musculoskeletal tissue homeostasis and recovery through the direct modulation of systemic inflammation, the immune response and the nociceptive pain.

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

Pro Football Sports Science Research Update

Recovery of the autonomic nervous system following football training among Division 1 collegiate football athletes: The influence of intensity and time. Heliyon, e18125, 2023.

Abstract - The autonomic nervous system (ANS) is profoundly affected by high intensity exercise. However, evidence is less clear on ANS recovery and function following prolonged bouts of high intensity exercise, especially in non-endurance athletes. Therefore, this study aimed to investigate the relationships between duration and intensity of acute exercise training sessions and ANS recovery and function in Division I football athletes. Fifty, male football athletes were included in this study. Subjects participated in 135 days of exercise training sessions throughout the 25-week season and wore armband monitors (Warfighter Monitor, Tiger Tech Solutions) equipped with electrocardiography capabilities. Intensity was measured via heart rate (HR) during an 'active state', defined as HR ≥ 85 bpm. Further, data-driven intensity thresholds were used and included HR < 140 bpm, HR < 150 bpm, HR < 160 bpm, HR ≥ 140 bpm, HR ≥ 150 bpm and HR ≥ 160 bpm. Baseline HR and HR recovery were measured and represented ANS recovery and function 24h post-exercise. Linear regression models assessed the relationships between time spent at the identified intensity thresholds and ANS recovery and function 24h post-exercise. Statistical significance set at $\alpha < 0.05$. Athletes participated in 128 training sessions, totaling 2735 data points analyzed. Subjects were predominantly non-Hispanic black (66.0%), aged 21.2 (±1.5) years and average body mass index of 29.2 (4.7) $kg\cdot(m^2)-1$. For baseline HR, statistically significant associations between duration and next-day ANS recovery were observed at HR < 140 bpm ($\beta = -0.08 \pm 0.02$, $R^2 = 0.31$, $p < 0.001$), HR above 150 and 160 bpm intensity thresholds ($\beta = 0.25 \pm 0.02$, $R^2 = 0.69$, $p < 0.0000$ and $\beta = 0.59 \pm 0.06$, $R^2 = 0.71$, $p < 0.0000$). Similar associations were observed for HR recovery: HR < 140 bpm ($\beta = 0.15 \pm 0.03$, $R^2 = 0.43$, $p < 0.0000$) and HR above 150 and 160 bpm ($\beta = -0.33 \pm 0.03$, $R^2 = 0.73$, $p < 0.0000$ and $\beta = -0.80 \pm 0.06$, $R^2 = 0.71$, $p < 0.0000$). The strengths of these associations increased with increasing intensity, HR ≥150 and 160 bpm (baseline HR: β range = 0.25 vs 0.59, R^2: 0.69 vs 0.71 and HR recovery: β range = − 0.33 vs − 0.80, R^2 = 0.73 vs 0.77). Time spent in lower intensity thresholds, elicited weaker associations with ANS recovery and function 24h post-exercise, with statistical significance observed only at HR < 140 bpm ($\beta = -0.08 \pm 0.02$, $R^2 = 0.31$, $p < 0.001$). The findings of this study showed that ANS recovery and function following prolonged high intensity exercise remains impaired for more than 24h. Strength and conditioning coaches should consider shorter bouts of strenuous exercise and extending recovery periods within and between exercise training sessions.

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

Pro Football Sports Science Research Update

Key Points

• The autonomic nervous system may require more than 24h of recovery following prolonged bouts of high-intensity exercise in collegiate football athletes.

• The "apparent restoration" of the autonomic nervous system 24h following, prolonged less strenuous exercise may be indicative of full recovery or parasympathetic hyperactivity.

• Strength and conditioning coaches must strategically design exercise training programs to allow for sufficient recovery of the autonomic nervous system, especially following prolonged bouts of high-intensity exercise.

Risk of two sport-related concussions in the same year: Is the second concussion worse? Clinical Journal of Sport Medicine, 34: 38-43, 2024.

Objectives: (1) Evaluate the frequency of same-year, repeat concussions; (2) assess predictors of sustaining a repeat concussion; and (3) compare outcomes of athletes with repeat concussions with athletes with single concussion.

Design: A retrospective, case–control study.

Setting: Regional sports concussion center.

Patients: Adolescents sustaining a sport-related concussions (SRC) from November 2017 to October 2020.

Independent Variables: Participants were dichotomized into 2 groups: (1) athletes with a single concussion; and (2) athletes with repeat concussions.

Main Outcome Measures: Between group and within group analyses were completed to look for differences in demographics, personal and family history, concussion history, and recovery metrics between the 2 groups.

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

Pro Football Sports Science Research Update

Results: Of 834 athletes with an SRC, 56 (6.7%) sustained a repeat concussion and 778 (93.3%) had a single concussion. Between group: Personal history of migraines (19.6% vs 9.5%, $x^2 = 5.795$, P = 0.02), family history of migraines (37.5% vs 24.5%, $x^2 = 4.621$, P = 0.03), and family history of psychiatric disorders (25% vs 13.1%, $x^2 = 6.224$, P = 0.01) were significant predictors of sustaining a repeat concussion. Within group: Among those with a repeat concussion, initial symptom severity was greater (Z 5 22.422; P 5 0.02) during the repeat concussion and amnesia was more common (x2 5 4.775, P 5 0.03) after the initial concussion.

Conclusions: In a single-center study of 834 athletes, 6.7% suffered a same-year, repeat concussion. Risk factors included personal/family migraine history and family psychiatric history. For athletes with repeat concussions, initial symptom score was higher after the second concussion, yet amnesia was more common after the initial concussion.

Specific collagen peptides increase adaptations of patellar tendon morphology following 14-weeks of high-load resistance training: A randomized-controlled trial. European Journal of Sport Science, Vol. 23 (12): 2329-2339, 2023.

Abstract - The purpose of this study was to investigate the effect of a supplementation with specific collagen peptides (SCP) combined with resistance training (RT) on changes in structural properties of the patellar tendon. Furthermore, tendon stiffness as well as maximal voluntary knee extension strength and cross-sectional area (CSA) of the rectus femoris muscle were assessed. In a randomized, placebo-controlled study, 50 healthy, moderately active male participants completed a 14-week resistance training program with three weekly sessions (70–85% of 1 repetition maximum [1RM]) for the knee extensors. While the SCP group received 5g of specific collagen peptides daily, the other group received the same amount of a placebo (PLA) supplement. The SCP supplementation led to a significant greater ($p < 0.05$) increase in patellar tendon CSA compared with the PLA group at 60% and 70% of the patellar tendon length starting from the proximal insertion. Both groups increased tendon stiffness ($p < 0.01$), muscle CSA ($p < 0.05$) and muscular strength ($p < 0.001$) throughout the intervention without significant differences between the groups. The current study shows that in healthy, moderately active men, supplementation of SCP in combination with RT leads to greater increase in patellar tendon CSA than RT

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

alone. Since underlying mechanisms of tendon hypertrophy are currently unknown, further studies should investigate potential mechanisms causing the increased morphology adaptions following SCP supplementation.

Highlights

- A daily supplementation of 5 g of specific collagen peptides during 14 weeks of high-load resistance training increase patellar tendon hypertrophy compared to the same training regimen and placebo.
- The resistance training-induced CSA increase, which was most pronounced on proximal and medial patellar tendon sites, is uniformly potentiated along the entire tendon length by supplementation.
- Patellar tendon stiffness, CSA of the rectus femoris muscle and maximal voluntary knee extension strength increase due to training independently from supplementation.
- Increased tendon CSA as a result of a stimulating effect of the supplementation with specific collagen peptides on collagen synthesis might be able to decrease tendon stress and support tendon healing.

Sports injuries and game stakes: Concussions in the National Football League. Economic Inquiry, Ahead of Print: 1-19, 2023.

Abstract - The National Football League's regular-season games are not of equal importance: some games loom larger than others for determining a team's chance to qualify for the playoffs. We develop an incentive-based measure of the impact of winning a game on a team's qualification probability to study the relationship between stakes and injuries. We find teams are 24 percentage points more likely to suffer concussions in games where a win secures one team a playoff berth. This is the first evidence to support the risk-escalation hypothesis that injuries increase with a competition's stakes. We then discuss implications for sports injury prevention.

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

Pro Football Sports Science Research Update

Strength and power thresholds to identify high and low linear sprint speed performers in collegiate American football players. Journal of Strength and Conditioning Research, Ahead of Print: 1-6, 2023.

Abstract - Lower-body strength and power are commonly measured performance qualities across a number of sports. In recent years, more attention has been given to relationships, primarily between lower-body strength and linear speed performance. While still limited, evidence is in agreement that lower-body strength positively contributes to linear speed performance. However, what is less well understood is if there comes a point in an athlete's development, at which, further working on increasing maximal strength may not fully compliment additional gains in speed performance. Within this study, authors aimed to provide practitioners with lower-body strength and power thresholds that can discriminate between slow and fast performers, within a group of collegiate American football players. The sample was further divided into a high-body and low-body weight group, and authors hypothesized that by using logistic regression, supplemented with receiver operator curve analyses, optimal cut-off points (i.e., relative lower-body strength thresholds) that are able to significantly discriminate between slow and fast linear speed performers may be identified. Findings indicate that optimal cut-off scores differed between the groups of athletes, as well as the lower body strength and power tests. All models were able to significantly distinguish between slower and faster performers, and area under the curve values ranged from 0.695 to 0.903. Although thresholds will likely vary based on factors such as sex, training age, and sport, findings from this investigation may be used to benchmark athletes and to further individualize training aimed at improving linear speed performance.

Practical Applications - This investigation puts forth different relative lower-body strength and power thresholds aimed at classifying athletes into high and low linear speed performance groups. Thresholds and model summary statistics differed based on the respective physical performance assessment and the group to which they were applied (high vs. low body mass groups). Within the high body mass group, vertical jump performance (AUC = 0.903), broad jump performance (AUC = 0.883), and relative hang clean strength (AUC = 0.848) showed the highest predictive ability regarding correctly classifying athletes into respective linear speed performance groups. These results reflect proficiency within movements involving higher rates of force development and a rapid hip extension within the group of faster athletes. Strength and conditioning practitioners may find this insightful regarding their selection of exercises enhancing such qualities. Within the low body mass group, vertical jump performance (AUC

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

= 0.779), broad jump performance (AUC = 753), and relative squat strength (AUC = 0.74) showed the highest predictive ability regarding correctly classifying athletes into respective linear speed performance groups. In this group, the multifactorial nature of speed development was reflected within the lower AUC values and higher model specificity, which may be acknowledged by coaches training populations with a wide range of anthropometric features. Interestingly, in both groups, the vertical jump exercise showed the highest predictive ability of correctly classifying athletes into respective speed groups. Strength and conditioning practitioners may use the vertical jump as a monitoring tool when training for lower body power to enhance sprint speed. Pending further longitudinal investigations of how increases in strength and changes in speed complement each other, methods and findings from this study may be used to benchmark athletes and to further individualize training implementations aimed at improving linear sprint speed performance through targeted interventions within the realms of strength and conditioning. Caution is advised when trying to generalize established thresholds past the population used within this study. Thresholds and findings will most likely vary based on factors such as sex, training age, sport, and measurement devices used.

The acute and chronic effects of implementing velocity loss thresholds during resistance training: A systematic review, meta-analysis and critical evaluation of the literature. Sports Medicine, 53(1): 177-214, 2023.

Background: Velocity loss (VL) experienced in a set during resistance training is often monitored to control training volume and quantify acute fatigue responses. Accordingly, various VL thresholds are used to prescribe resistance training and target different training adaptations. However, there are inconsistencies in the current body of evidence regarding the magnitude of the acute and chronic responses to the amount of VL experienced during resistance training.

Objective: The aim of this systematic review was to (1) evaluate the acute training volume, neuromuscular, metabolic, and perceptual responses to the amount of VL experienced during resistance training; (2) synthesize the available evidence on the chronic effects of different VL thresholds on training adaptations; and (3) provide an overview of the factors that might differentially influence the magnitude of specific acute and chronic responses to VL during resistance training.

Pro Football Sports Science Research Update

Methods: This review was performed using the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines. Five databases were searched, and studies were included if they were written in English, prescribed resistance training using VL, and evaluated at least one (1) acute training volume, neuromuscular, metabolic, or perceptual response or (2) training adaptation. Risk of bias was assessed using a modified Cochrane Collaboration's tool for assessing the risk of bias in randomized trials. Multilevel and multivariate meta-regressions were performed where possible.

Results: Eighteen acute and 19 longitudinal studies met the inclusion criteria, of which only one had more than one risk of bias item assessed as high risk. Based on the included acute studies, it seems that the number of repetitions per set, blood lactate concentration, and rating of perceived exertion generally increase, while countermovement jump height, running sprint times, and velocity against fixed loads generally decrease as VL increases. However, the magnitude of these effects seems to be influenced, among other factors, by the exercise and load used. Regarding training adaptations, VL experienced during resistance training did not influence muscle strength and endurance gains. Increases in VL were associated with increases in hypertrophy (b = 0.006; 95% confidence interval [CI] 0.001, 0.012), but negatively affected countermovement jump (b = - 0.040; 95% CI - 0.079, - 0.001), sprint (b = 0.001; 95% CI 0.001, 0.002), and velocity against submaximal load performance (b = - 0.018; 95% CI - 0.029, - 0.006).

Conclusions: A graded relationship exists between VL experienced during a set and acute training volume, neuromuscular, metabolic, and perceptual responses to resistance training. However, choice of exercise, load, and individual trainee characteristics (e.g., training history) seem to modulate these relationships. The choice of VL threshold does not seem to affect strength and muscle endurance gains whereas higher VL thresholds are superior for enhancing hypertrophy, and lower VL thresholds are superior for jumping, sprinting, and velocity against submaximal loads performance.

Key Points

- A graded relationship exists between velocity loss (VL) experienced during a set and acute training volume, neuromuscular, metabolic, and perceptual responses to resistance training with factors such as type of exercise, loads used, and individual characteristics of a trainee seeming to modulate these relationships.

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

Pro Football Sports Science Research Update

- Factors that can specifically affect the consistency of VL determination include reference repetitions, velocity variables (e.g., mean or peak), and criteria for set termination after VL has been exceeded, all of which should be considered when implementing VL in practice.
- The amount of VL experienced during resistance training does not seem to affect strength and muscle endurance gains whereas higher VL may be superior when the aim is to induce hypertrophy. Allowing only low to moderate VL during resistance training seems to be a viable strategy for optimizing jumping, sprinting, and velocity against submaximal loads performance.
- As higher VL experienced during resistance training could interfere with the ability to rapidly produce force, cause a reduction in the expression of fast-twitch muscle fibers, and prolong recovery from resistance training, low to moderate VL could be recommended to optimize strength and power training adaptations as well as the performance of sport-specific tasks. However, if hypertrophy is also the goal, more of the prescribed sets could utilize moderate VLs, or more total sets with low to moderate VL could be performed.

The ALR-RSI score can be used to evaluate psychological readiness to return to sport after acute Achilles tendon tear. Knee Surgery, Sports Traumatology, Arthroscopy. 31: 4961-4968, 2023.

Purpose: The return to sport is one of the main goals following Achilles tendon tear repair. Several psychological factors influence the return to sport after a sports injury. The traditional tools to assess the return to sport do not take into account psychological factors. The ankle ligament reconstruction-return to sport injury (ALR-RSI), validated for ankle instability, is a score to evaluate psychological readiness to return to sport. The goal of this study was to validate the ALR-RSI score for the assessment of the readiness to return to sport after Achilles tendon repair.

Methods: The ALR-RSI score, adapted from the anterior cruciate ligament-return to sport injury (ACL-RSI) score used following knee ligament reconstruction, was validated according to the international COSMIN methodology. Patients operated for Achilles tendon repair responded to the questionnaire during the rehabilitation period. The EFAS, FAAM and VISA-A scores were used as reference questionnaires.

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

Results: A total of 50 patients were included. The ALR-RSI score was strongly (r > 0.5) correlated to the EFAS score: r = 0.68 [0.50–0.80] the FAMM sport score: r = 0.7 [0.52–0.84] the FAAM AVQ score (r = 0.6 [0.35–0.78]), and the VISA-A score (r = 0.54 [0.26–0.76]). The discriminant validity was good with the ALR-RSI, which was significantly lower in the patients that did not return to sport: 60.7 (40–81.4) compared to those that did: 83.2 (64.3–100) p = 0.001. Reproducibility was excellent with an intra-class correlation coefficient ρ of 0.99 [097–1.00]. The internal consistency was excellent (alpha coefficient = 0.95).

Conclusion: The ALR-RSI score provides a valid, reproducible assessment of the psychological readiness to return to sport in patients who undergo surgical Achilles tendon suture repair.

Level of evidence: III.

The impact of eccentric or isometric training on strength, architecture, and sprinting across an Australian football season. Medicine and Science in Sports and Exercise, Ahead of Print: 1-45, 2023.

Purpose: Investigate impact of an isometric (ISO) or Nordic hamstring exercise (NHE) intervention, alongside a sprint training program on hamstring strength, architecture and sprinting performance in Australian Footballers.

Methods: Twenty-five male athletes undertook NHE (n = 13) or ISO (n = 12) training across a 38-week period (including pre-season and in-season). Biceps femoris long head (BFlh) architecture, isometric and eccentric knee flexor strength were assessed at baseline, end of pre-season (14-weeks) and at the conclusion of the intervention. Sprint times and force-velocity profiles were determined at baseline and at the end of pre-season.

Results: Following the intervention both groups had significant improvements in BFlh fascicle length (NHE: 1.16 cm, 95%CI 0.68 to 1.63 cm, d = 1.88, p < 0.001; ISO: 0.82 cm, 95%CI 0.57 to 1.06 cm, d = 1.70, p < 0.001), muscle thickness (NHE: 0.11 cm, 95%CI 0.01 to 0.21 cm, d = 0.51, p = 0.032; ISO: 0.21 cm, 95%CI 0.10 to 0.32 cm, d = 0.86, p = 0.002) and eccentric strength (NHE: 83 N, 95%CI 53 to 114 N, d = 1.79, p < 0.001; ISO: 83 N, 95%CI 17 to 151 N, d = 1.17, p = 0.018). Both groups also finished the

intervention weaker isometrically than they started (NHE: -45 N, 95%CI -81 to -8 N, d = -1.03, p = 0.022; ISO: -80 N, 95%CI -104 to -56 N, d = -3.35, p < 0.001). At the end of pre-season the NHE group had improved their 5 m sprint time by 3.3% (±2.0%) and their maximum horizontal velocity was 3% (±2.1) greater than the ISO group who saw no changes.

Conclusion: Both ISO and NHE training with a periodized sprinting program can increase BFlh fascicle length, thickness, and eccentric strength in Australian Footballers. NHE training also improves 5 m sprint time and maximum velocity. However, both interventions reduced isometric strength. These findings provide unique, contextually relevant insights into the adaptations possible in semi-professional athletes.

The multifaceted nature of recovery after exercise: A need for individualization. Sports Orthopaedics and Traumatology, Ahead of Print: 1-9, 2023.

Abstract - A proper balance between load and recovery is crucial to initiate (optimal) physiological adaptations and avoid overuse injuries. Recovery in volves physical and psychological aspects and can be enhanced through various recovery methods. The choice of a specific recovery method depends on factors like the type of activity, time until the next competition, and equipment accessibility, with considerations for the training phase and competition schedule playing a crucial role. Furthermore, different aspects of recovery follow varying individual and inter-individual time spans, making it challenging to apply group-based information to the individual level from single parameters (i.e., metabolic, neuromuscular, central, cardio-regulatory or performance-based parameters). Thus, there is a need for personalized approaches to optimize post-exercise recovery strategies. Combining group-based knowledge and repeatedly measured individual data may help practitioners to design tailored strategies.

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

Pro Football Sports Science Research Update

Tracking devices and physical performance analysis in team sports: A comprehensive framework for research-trends and future directions. Frontiers in Sports and Active Living, 5:1284086, 2023.

Background: Tracking devices, such as global (GPS) and local (LPS) positioning systems, combined with physiological measurements, have become reliable tools to characterize movement patterns, assessing the external load (EL), internal load (IL), fatigue, and performance of athletes in team sports. This scoping review aims to provide a comprehensive understanding of the applicability of tracking systems in physical performance analysis within team sports and the wellbeing of athletes based on research strategies and combined variables.

Methods: A systematic search was conducted in PubMed, Web of Knowledge, and Scopus databases according to PRISMA guidelines. The 79 studies that were reviewed met the following criteria: (1) contained relevant data regarding elite athletes' performance; (2) athletes' EL and IL; (3) were written in the English language; (4) were related only to team sports.

Results: The findings indicate that tracking technology has been engaged in several research areas, including performance analysis, training vs. match load management, injuries, and nutrition, through characterization and correlational studies. Metrics, primarily focused on kinematic and mechanical EL aspects, have been employed in combination with IL data to analyze the performance of athletes. However, the lack of an integrative model for the analysis and integration of EL and IL metrics within each team sport suggests an interesting direction for further research.

Conclusion: There is a need for coherence between the methods and the research goals on performance analysis. The development of a framework that guides experimental studies is highly recommended, particularly on manipulating metrics analyzed between training and match sessions, injury prevention, and nutrition. This will lead to the development of the most applied sports science research to improve the preparation and decision-making of athletes based on reliable data.

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

Pro Football Sports Science Research Update

Ultrasound shear wave seeds reduced following hamstring strain injury but not after returning to sport. Insights into Imaging, 15:7, 2024.

Objectives: The purpose of the study was to investigate differences in ultrasound shear wave speed (SWS) between uninjured and injured limbs following hamstring strain injury (HSI) at time of injury (TOI), return to sport (RTS), and 12 weeks after RTS (12wks).

Methods: This observational, prospective, cross-sectional design included male and female collegiate athletes who sustained an HSI. SWS imaging was performed at TOI, RTS, and 12wks with magnetic resonance imaging. SWS maps were acquired by a musculoskeletal-trained sonographer at the injury location of the injured limb and location-matched on the contralateral limb. The average SWS from three 5 mm diameter Q-boxes on each limb were used for analysis. A linear mixed effects model was performed to determine differences in SWS between limbs across the study time points.

Results: SWS was lower in the injured limb compared to the contralateral limb at TOI (uninjured – injured limb difference: 0.23 [0.05, 0.41] m/s, p=0.006). No between-limb differences in SWS were observed at RTS (0.15 [-0.05, 0.36] m/s, p=0.23) or 12wks (-0.11 [-0.41, 0.18] m/s, p=0.84).

Conclusions: The SWS in the injured limb of collegiate athletes after HSI was lower compared to the uninjured limb at TOI but not at RTS or 12 weeks after RTS.

Critical relevance statement: Hamstring strain injury with structural disruption can be detected by lower injured limb shear wave speed compared to the uninjured limb. Lack of between-limb differences at return to sport may demonstrate changes consistent with healing. Shear wave speed may complement traditional ultrasound or MRI for monitoring muscle injury.

Key points

• Ultrasound shear wave speed can non-invasively measure tissue elasticity in muscle injury locations.

• Injured limb time of injury shear wave speeds were lower versus uninjured limb but not thereafter.

• Null return to sport shear wave speed differences may correspond to structural changes associated with healing.

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

Pro Football Sports Science Research Update

• Shear wave speed may provide quantitative measures for monitoring muscle elasticity during recovery.

Unlocking the healing potential: Cannabinoids in spine surgery for pain relief and recovery. JB&JS Reviews, 11(11):e23.00141, 2023.

Abstract

- Cannabinoids, such as D9-tetrahydrocannabinol and cannabidiol, interact with endocannabinoid receptors in the central nervous system and immune system, potentially offering pain relief. The entourage effect, resulting from the interaction of multiple cannabis components, may enhance therapeutic impact and efficacy, making them promising candidates for exploring pain relief in spine operations, known to be among the most painful operative procedures.
- The use of cannabinoids in pain management requires careful consideration of safety, including their cognitive and psychomotor effects, potential cardiovascular risks, risk of dependence, mental health implications, and drug interactions.
- Few studies have analyzed cannabinoid use in relation to spine surgery, with variable results reported, indicating possible effects on reoperation rates, mortality, complications, postoperative opioid use, and length of hospital stay.
- Current knowledge gaps exist in the understanding of cannabinoid effects on spine surgery, including the exploration of different administration routes, timing, dosage, and specific outcomes. In addition, mechanistic explanations for the observed results are lacking.
- Ethical considerations related to informed consent, medical expertise, societal impact, and legal compliance must also be thoroughly addressed when considering the utilization of cannabinoids in spinal pathologies and back pain treatment.

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

Pro Football Sports Science Research Update

Use of extracorporeal shockwave therapies for athletes and physically active individuals: A systematic review. British Journal of Sports Medicine, Ahead of Print: 1-12, 2024.

Objective: To determine the efficacy of extracorporeal shockwave therapy (ESWT) and investigate outcomes following the use of ESWT for athletes and physically active individuals.

Design: Systematic review.

Data sources: We followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses and searched four databases: PubMed (NLM), Embase (Elsevier), CINAHL Complete (EBSCO) and Web of Science (Clarivate).

Eligibility criteria for selecting studies: Included studies were randomized controlled trials, cohort and case-control studies, cases series and reports that evaluated outcomes following ESWT for athletes, physically active individuals and occupational groups requiring regular physical activity such as military cadets.

Results: 56 studies with 1874 athletes or physically active individuals were included. Using the Oxford level of evidence rating, included studies were 18 level I (32.1%), 3 level II (5.4%), 10 level III (17.9%), 13 level IV (23.2%) and 12 level V (21.4%). Based on the level I studies, ESWT may be effective alone in plantar fasciitis, lateral epicondylitis and proximal hamstring tendinopathy and as an adjunct to exercise treatment in medial tibial stress syndrome and osteitis pubis in athletes or physically active individuals. In most studies, athletes were allowed to continue activities and training and tolerated ESWT with minimal side effects.

Conclusion: ESWT may offer an efficacious treatment alone or as an adjunct to concurrent exercise therapy in selected sports-related injuries and without major adverse events. Further high-level research is needed to better define the role and clinical outcomes of ESWT.

WHAT IS ALREADY KNOWN

⇒ Extracorporeal shockwave therapy (ESWT) has been widely used for the management of common sports-related injuries both in physically active and sedentary patient cohorts.

INFINITE ATHLETE

To learn more about Infinite Athlete - http://infiniteathlete.ai/

Pro Football Sports Science Research Update

WHAT ARE THE NEW FINDINGS

⇒ Patellar tendinopathy is the most widely studied condition for athletes, but the use of ESWT presents conflicting results.

⇒ Most studies evaluating ESWT for athletes and physically active individuals have a low level of evidence.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ Additional high-level research is needed to better define the role and clinical outcomes of ESWT in athletes and physically active individuals

To learn more about Infinite Athlete - http://infiniteathlete.ai/